

Paper 2109 Unit 1

Selected question formula and method map

မေးခွန်းနံပါတ်မြင်တာနဲ့ သုံးရမည့် formula ကို အမြန်ရွေးနိုင်ရန် စုစည်းထားသော sheet ဖြစ်သည်။
Formula များကို သတ်မှတ်ထားသော vector arrow နှင့် unit-vector hat notation ဖြင့် ရေးထားသည်။

Core formulas to remember

$$\hat{\mathbf{i}} \cdot \hat{\mathbf{i}} = \hat{\mathbf{j}} \cdot \hat{\mathbf{j}} = \hat{\mathbf{k}} \cdot \hat{\mathbf{k}} = 1, \quad \hat{\mathbf{i}} \cdot \hat{\mathbf{j}} = \hat{\mathbf{j}} \cdot \hat{\mathbf{k}} = \hat{\mathbf{k}} \cdot \hat{\mathbf{i}} = 0$$

$$|\vec{a}| = \sqrt{\vec{a} \cdot \vec{a}} = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3, \quad \vec{a} \cdot \vec{b} = 0 \Rightarrow \text{perpendicular}$$

$$\hat{\mathbf{i}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}}, \quad \hat{\mathbf{j}} \times \hat{\mathbf{k}} = \hat{\mathbf{i}}, \quad \vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$$

Vector arrow နှင့် unit-vector hat များကို formula renderer ဖြင့် သီးခြားတိတိကျကျ ထည့်ထားသည်။

No. 1 - Ex. 1.1 Magnitude identity

$$\vec{a} = a_1 \hat{\mathbf{i}} + a_2 \hat{\mathbf{j}} + a_3 \hat{\mathbf{k}}$$

$$\vec{a} \cdot \vec{a} = a_1^2 + a_2^2 + a_3^2$$

$$|\vec{a}| = \sqrt{\vec{a} \cdot \vec{a}} = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

မှတ်ရန် - magnitude = self dot product ၏ square root

No. 2 - Model Question No. 2 Angle ABC

$$\overrightarrow{BA} = \vec{a} - \vec{b}, \quad \overrightarrow{BC} = \vec{c} - \vec{b}$$

$$\overrightarrow{BA} \cdot \overrightarrow{BC} = 0 \Rightarrow \angle ABC = 90^\circ$$

B ထောင့်ဖြစ်သောကြောင့် vector နှစ်ခုလုံးကို B မှစရမည်။

No. 3 - S.A. 1.1 (1) Vector identity proof

$$(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = ((\vec{a} \times \vec{b}) \times \vec{c}) \cdot \vec{d}$$

$$(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$$

$$(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = (\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{d}) - (\vec{a} \cdot \vec{d})(\vec{b} \cdot \vec{c})$$

အဓိက - scalar triple product cyclic property နှင့် vector triple product identity ကို ဆက်သုံး။

No. 4 - S.A. 1.1 (2)(i) Scalar triple product

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

အရင် b vector cross c ရှာပြီး a vector နှင့် dot လုပ်နိုင်သည်။ Final answer သည် scalar ဖြစ်သည်။

No. 5 - S.A. 1.1 (2)(ii) Vector triple product

$$(\vec{a} \times \vec{b}) \times \vec{c} = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{a}(\vec{b} \cdot \vec{c})$$

မှတ်ရန် - middle vector b ကို (a dot c) နှင့်မြှောက်၊ first vector a ကို (b dot c) နှင့်မြှောက်ပြီး နုတ်။

No. 6 - Ex. 1.2 Double cross product

$$\vec{p} = \vec{a} \times \vec{b} \Rightarrow \vec{p} \times \vec{c}$$

$$(\vec{a} \times \vec{b}) \times \vec{c} = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{a}(\vec{b} \cdot \vec{c})$$

Cross-product determinant နှစ်ဆင့်သုံးနိုင်သည်။ (a cross b) cross c နှင့် a cross (b cross c) မတူပါ။

No. 7 - Question No. 3 Directional derivative

$$\nabla \phi = \text{grad } \phi = \phi_x \hat{\mathbf{i}} + \phi_y \hat{\mathbf{j}} + \phi_z \hat{\mathbf{k}}$$

$$|\vec{d}| = \sqrt{d_1^2 + d_2^2 + d_3^2}, \quad \hat{\mathbf{u}} = \frac{\vec{d}}{|\vec{d}|}$$

$$D_{\hat{\mathbf{u}}} \phi = \hat{\mathbf{u}} \cdot \nabla \phi$$

အမြန်မှတ်ရန် - unit direction dot gradient. Direction vector d ကို unit vector u ပြောင်းပြီးမှ gradient နှင့် dot လုပ်။

No. 8 - Question No. 1 Harmonic function

$$r = \sqrt{x^2 + y^2 + z^2}, \quad \phi = \frac{1}{r}$$

$$\nabla^2 \phi = \phi_{xx} + \phi_{yy} + \phi_{zz}$$

$$\phi_{xx} = -\frac{1}{r^3} + \frac{3x^2}{r^5}, \quad \phi_{yy} = -\frac{1}{r^3} + \frac{3y^2}{r^5}, \quad \phi_{zz} = -\frac{1}{r^3} + \frac{3z^2}{r^5}$$

$$\nabla^2 \left(\frac{1}{r} \right) = -\frac{3}{r^3} + \frac{3(x^2 + y^2 + z^2)}{r^5} = 0, \quad r \neq 0$$

Harmonic ဖြစ်ကြောင်းပြရန် Laplacian ကို 0 ဖြစ်ကြောင်းပြရမည်။ Origin r=0 တွင် 1/r မသတ်မှတ်နိုင်။

No. 9 - Question No. 2 Closed circuit

$$\nabla \left(\frac{\phi^2}{2} \right) = \phi \nabla \phi$$

$$\oint_C \phi \nabla \phi \cdot d\vec{s} = \oint_C \nabla \left(\frac{\phi^2}{2} \right) \cdot d\vec{s} = 0$$

Closed path တွင် start = end ဖြစ်သောကြောင့် gradient field ၏ line integral သည် 0 ဖြစ်သည်။

No. 10 - Question No. 3 Gradient

$$\nabla \phi = \phi_x \hat{\mathbf{i}} + \phi_y \hat{\mathbf{j}} + \phi_z \hat{\mathbf{k}}$$

$$\hat{\mathbf{u}} = \frac{\vec{d}}{|\vec{d}|}, \quad D_{\hat{\mathbf{u}}} \phi = \nabla \phi \cdot \hat{\mathbf{u}}$$

Point တွင် gradient ကို အရင်တန်ဖိုးချ၊ direction ကို unit ပြောင်း၊ နောက်ဆုံး dot product လုပ်။

Question number to formula map

No. 1 - Ex. 1.1

$$|\vec{a}| = \sqrt{\vec{a} \cdot \vec{a}}$$

No. 2 - Angle ABC

$$\overrightarrow{BA} = \vec{a} - \vec{b}, \quad \overrightarrow{BC} = \vec{c} - \vec{b}, \quad \overrightarrow{BA} \cdot \overrightarrow{BC} = 0$$

No. 3 - Identity proof

$$(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = (\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{d}) - (\vec{a} \cdot \vec{d})(\vec{b} \cdot \vec{c})$$

No. 4 - Scalar triple

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = \text{determinant}$$

No. 5 - Vector triple

$$(\vec{a} \times \vec{b}) \times \vec{c} = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{a}(\vec{b} \cdot \vec{c})$$

No. 6 - Double cross

$$\vec{p} = \vec{a} \times \vec{b} \Rightarrow \vec{p} \times \vec{c}$$

No. 7 - Directional derivative

$$D_{\hat{\mathbf{u}}} \phi = \hat{\mathbf{u}} \cdot \nabla \phi \quad (\text{unit direction dot gradient})$$

No. 8 - Harmonic

$$\nabla^2(1/r) = 0$$

No. 9 - Closed circuit

$$\oint_C \phi \nabla \phi \cdot d\vec{s} = 0$$

No. 10 - Gradient

$$\nabla \phi = \phi_x \hat{\mathbf{i}} + \phi_y \hat{\mathbf{j}} + \phi_z \hat{\mathbf{k}}, \quad D_{\hat{\mathbf{u}}} \phi = \nabla \phi \cdot \hat{\mathbf{u}}$$

Last minute memory chain

1. Magnitude : $|\vec{a}| = \sqrt{\vec{a} \cdot \vec{a}}$

2. Right angle : $\overrightarrow{BA} = \vec{a} - \vec{b}, \overrightarrow{BC} = \vec{c} - \vec{b}, \overrightarrow{BA} \cdot \overrightarrow{BC} = 0$

3. Scalar triple : $\vec{a} \cdot (\vec{b} \times \vec{c})$

4. Vector triple : $(\vec{a} \times \vec{b}) \times \vec{c}$

5. Directional : $\hat{\mathbf{u}} \cdot \nabla \phi$

6. Harmonic : $\nabla^2 \phi = 0$

7. Closed circuit : $\oint \nabla \left(\frac{\phi^2}{2} \right) \cdot d\vec{s} = 0$

8. Gradient : $\nabla \phi = \phi_x \hat{\mathbf{i}} + \phi_y \hat{\mathbf{j}} + \phi_z \hat{\mathbf{k}}$